Q1) $\int \cosh(3x-2)dx =$

a) $\frac{-1}{2}\sinh(3x-2)+c$ b) $\frac{1}{2}\sinh(3x-2)+c$ c) $\frac{-1}{3}\sinh(3x-2)+c$ d) $\frac{1}{3}\sinh(3x-2)+c$

Q2) $\lim_{x\to 0}\frac{e^{3x}-1}{\sin 2x} =$

a) $+\infty$ b) 0 c) $\frac{3}{2}$ d) $\frac{2}{3}$

Q3) To evaluate the integral $\int\sqrt{9-4x^2}\,dx$, The best trigonometric substitution is

a) $x=\frac{2}{3}\tan\theta$ b) $x=\frac{2}{3}\sec\theta$ c) $x=\frac{3}{2}\sin\theta$ d) $x=\frac{2}{3}\sin\theta$

Q4) $\int\frac{dx}{x\sqrt{x^2-9}}$

a) $\sin^{-1}\left|\frac{x}{3}\right|+c$ b) $\frac{1}{3}\sec^{-1}\left|\frac{x}{3}\right|+c$ c) $\sec^{-1}\left|\frac{x}{3}\right|+c$ d) $\frac{1}{3}\sin^{-1}\left|\frac{x}{3}\right|+c$

Q5) *The partial fraction decomposition for the integral* $\int\frac{x^2-5}{(x^2+3x)(x^2+9x)}dx$ is

a) $\frac{A_1}{x}+\frac{A_2}{x+3}+\frac{A_3x+A_4}{x^2+9}+\frac{A_5}{x}$ b) $\frac{A_1}{x}+\frac{A_2}{x^2}+\frac{A_3}{x+3}+\frac{A_4}{(x-3)^2}+\frac{A_5}{x-3}$

c) $\frac{A_1}{x}+\frac{A_2}{x^2}+\frac{A_3}{x-3}+\frac{A_4}{(x-3)^2}+\frac{A_5}{x+3}$ d) $\frac{A_1}{x}+\frac{A_2}{x^2}+\frac{A_3}{x+3}+\frac{A_4x+A_5}{x^2+9}$

Q6) One of the following integrals is improper

a) $\int_4^5\frac{dx}{\ln(x-3)}$ b) $\int_1^e\frac{\ln x}{x}dx$ c) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos x\,dx$ d) $\int_1^2(x-2)^2$

Q7) If $y=x^{3x}$, then $\frac{dy}{dx}=$

a) $(3x)x^{3x-1}$ b) $(3+\ln x)x^{3x}$ c) $(3+3\ln x)x^{3x}$ d) $3x^{3x-1}$

Q8) $\int 3x\sec^2 x\,dx$

a) $3x\tan x+c$ b) $3x\ln|\sec x|+c$ c) $3x\tan x+\ln|\sec x|+c$ d) $3x\tan x-3\ln|\sec x|+c$

Q9) The best substitution to integrate $\int\cos^4 x\sin^5 x\,dx$ is

a) $u=\sin x$ b) $u=\cos x$ c) $u=\sin^5 x$ d) $u=\cos^4 x$

Q10) $\int\cos 2x\sin 7x\,dx$

a) $\left(\frac{\cos 9x}{9}+\frac{\cos 5x}{5}\right)+c$ b) $-\left(\frac{\cos 9x}{18}+\frac{\cos 5x}{10}\right)+c$

c) $\left(\frac{\cos 9x}{18}-\frac{\cos 5x}{10}\right)+c$ d) $\left(\frac{\cos 9x}{18}+\frac{\cos 5x}{10}\right)+c$

Q11) The sequence $\left\{\frac{3n-1}{5^n}\right\}_{n=1}^{\infty}$ converges to

a) $\frac{3}{5}$ b) $\frac{3}{5\ln 5}$ c) $3\ln 5$ d) 0

Q12) The sequence $\{4,3,2,1,0,1,2,3,4,5,\ldots\}$

a) Strictly eventually decreasing b) Strictly eventually increasing

c) Strictly increasing d) Strictly decreasing

Q13) The series $\sum_{k=1}^{\infty}\frac{1}{k^{p-4}}$ converges if

a) $p>5$ b) $p>4$ c) $p>3$ d) $p<4$

Q14) Maclaurin series for $f(x)=e^{2x}-e^{-2x}$ is

a) $\sum_{k=0}^{\infty}\frac{(2x)^k}{2k!}$ b) $\sum_{k=0}^{\infty}\frac{(-2x)^k}{k!}$ c) $\sum_{k=0}^{\infty}\frac{4^{k+1}x^{2k+1}}{(2k+1)!}$ d) $\sum_{k=0}^{\infty}\frac{2x^k}{k!}$

Q15) The sum of the series $\sum_{k=1}^{\infty}\frac{1}{k+3}-\frac{1}{k+4}+\left(\frac{-1}{2}\right)^{k+1}$ is

a) $\frac{3}{4}$ b) $\frac{5}{12}$ c) $\frac{1}{12}$ d) $\frac{5}{4}$

Q16) One of the following series converges

a) $\sum_{k=1}^{\infty}\frac{(-1)^k}{k}$ b) $\sum_{k=1}^{\infty}(k+1)!$ c) $\sum_{k=1}^{\infty}\frac{k+1}{k^2+4}$ d) $\sum_{k=1}^{\infty}\ln k$

Q17) Using ratio test, the value of a that makes the series $\sum_{k=1}^{\infty}\frac{3k}{a^k}$, $a>0$ converge

a) $a<1$ b) $a>1$ c) $a>3$ d) $a>\frac{1}{3}$

Q18) The series $\sum_{k=1}^{\infty}\left(\frac{ak+1}{2k+7}\right)^k$ converges if

a) $0<a<2$ b) $a>2$ c) $0<a<\frac{1}{2}$ d) $0<a<1$

Q19) The series $\sum_{k=1}^{\infty}\frac{2^{2k+1}}{(k-1)!}=$

a) $\frac{1}{2}e^4$ b) $2e^2$ c) $8e^4$ d) $2e^4$

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a) $\frac{-5}{6}$ b) $\frac{\pi}{4}-1$ c) $\ln 3$ d) $\sin 1$

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c) $\frac{A_1}{x}+\frac{A_2}{x^2}+\frac{A_3}{x-3}+\frac{A_4}{(x-3)^2}+\frac{A_5}{x+3}$ d) $\frac{A_1}{x}+\frac{A_2}{x^2}+\frac{A_3}{x+3}+\frac{A_4x+A_5}{x^2+9}$

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a) $\frac{-5}{6}$ b) $\frac{\pi}{4}-1$ c) $\ln 3$ d) $\sin 1$

Subject / /

1) $\int \cosh(3x-2)\,dx = \frac{1}{3}\sinh(3x-2)+C$ (D)

2) $\lim_{x\to 0} \frac{e^{3x}-1}{\sin 2x} \left(\frac{0}{0}\right) \overset{L.R}{=} \lim_{x\to 0} \frac{3e^{3x}}{2\cos 2x} = \frac{3}{2}$ (C)

3) $\int \sqrt{9-4x^2}\,dx \longrightarrow 3\int \sqrt{1-\left(\frac{2x}{3}\right)^2}\,dx$

sub: $\sin\theta = \frac{2x}{3} \longrightarrow x = \frac{3}{2}\sin\theta$ (C)

4) $\int \frac{dx}{x\sqrt{x^2-9}} = \frac{1}{3}\int \frac{dx}{x\sqrt{\left(\frac{x}{3}\right)^2-1}}$

$\sec\theta = \frac{x}{3} \longrightarrow 3\sec\theta = x$

$3\sec\theta\tan\theta\,d\theta = dx$

$= \frac{1}{3}\int \frac{3\sec\theta\tan\theta}{3\sec\theta\sqrt{\sec^2\theta-1}}\,d\theta = \frac{1}{3}\int \frac{\sec\theta\tan\theta}{\sec\theta\tan\theta}\,d\theta$

$= \frac{1}{3}\theta + C = \frac{1}{3}\sec^{-1}\left(\frac{x}{3}\right)+C$ (B)

5) $\int \frac{x^2-5}{(x^2+3x)(x^3+9x)}\,dx$

$\frac{x^2-5}{x(x+3)x(x^2+9)} = \frac{x^2-5}{x^2(x+3)(x^2+9)} = \frac{A_1}{x} + \frac{A_2}{x^2} + \frac{A_3}{x+3} + \frac{A_4x+A_5}{x^2+9}$

(d)

Subject ..... / /

6) which of the following integrals is improper :-

A. $\int_3^6 \frac{dx}{\ln(x-3)}$ B. $\int_1^e \frac{\ln x}{x}\,dx$

(A)

C. $\int_{\frac{\pi}{2}}^{\pi} \cos x\,dx$ d. $\int_2^4 (x-2)^2\,dx$

7) $y = x^{3x}$, $y' = ??$

$\ln y = 3x \ln x$

$\frac{y'}{y} = 3x\frac{1}{x} + 3\ln x$

$y' = y(3 + 3\ln x) = (3 + 3\ln x)\, x^{3x}$ (C)

8) $\int 3x \sec^2 x\,dx$

D	I
u	dv
$3x$	$\sec^2 x$
3	$\tan x$
0	$-\ln\lvert\cos x\rvert$

$= 3x\tan x - 3\ln\lvert\cos x\rvert + C$

$= 3x\tan x - 3\ln\lvert\sec x\rvert + C$ (D)

9) $\int \cos^4 x \sin^3 x\,dx = \int \cos^4 x \underbrace{\sin^2 x}_{(1-\cos^2 x)} \underbrace{\sin x\,dx}_{\frac{d}{dx}\cos x}$

therefore, best sub. is $u = \cos x$.

Subject / /

10) $\int \cos 2x \sin 7x \, dx$

$= \int \sin 7x \cos 7x \, dx = \frac{1}{2} \int \sin(9x) + \sin(5x) \, dx$

$= -\frac{\cos(9x)}{18} - \frac{\cos 5x}{10} + C$

$= -\left(\frac{\cos 9x}{18} + \frac{\cos 5x}{10} \right) + C$ (D)

11) $\left\{ \frac{3n-1}{5^n} \right\}_{n=1}^{\infty}$ converges to ??

$\lim_{n \to \infty} \left(\frac{3n-1}{5^n} \right) \overset{\frac{\infty}{\infty}}{\overset{L.R}{=}} \lim_{n \to \infty} \left(\frac{3}{5^n} \right) = 0$ (D)

12) The seq: $\{4, 3, 2, 1, 0, 1, 2, 3, 4, \dots\}$

منخفض بعدين يبدأ بالصفر ويزداد

Strictly eventually increasing.

Subject ..... / /

15) $\sum_{k=1}^{\infty} \frac{1}{k+3} - \frac{1}{k+4} + \left(-\frac{1}{2}\right)^{k+1}$

Telescoping:

$k=1$ $k=2$ $k=3$

$\frac{1}{4} - \frac{1}{5} + \frac{1}{5} - \frac{1}{6} + \frac{1}{6} - \frac{1}{7} + \cdots$

$k=k$

$+ \frac{1}{k+3} - \frac{1}{k+4}$

$S = \lim_{k\to\infty} S_k = \frac{1}{4} - \lim_{k\to\infty} \frac{1}{k+4}$ (G)

$S_1 = 1/4$

Geometric:

$\sum_{k=1}^{\infty} \left(-\frac{1}{2}\right)^{k-1+2}$

$\left(-\frac{1}{2}\right)^2 \sum_{k=1}^{\infty} \left(-\frac{1}{2}\right)^{k-1}$

a r

$S = \frac{a}{1-r} = \frac{1/4}{1-\left(-\frac{1}{2}\right)} = \frac{1}{4} \times \left(\frac{2}{3}\right)$

$S_2 = 1/6$

$\therefore S = S_1 + S_2 = \frac{1}{4} + \frac{1}{6} = \frac{6+4}{24}$

$= \frac{10}{24} = \frac{2 \cdot 5}{2 \cdot 12} = \frac{5}{12}$ (B)

(16) which series converges

A) $\sum_{k=1}^{\infty} \frac{(-1)}{k}$ Alternating Harmonic series $\xrightarrow{A.S.T}$ ① $\lim_{k\to\infty} \frac{1}{k} = 0$ ✓

② s.t.c decreasing ✓

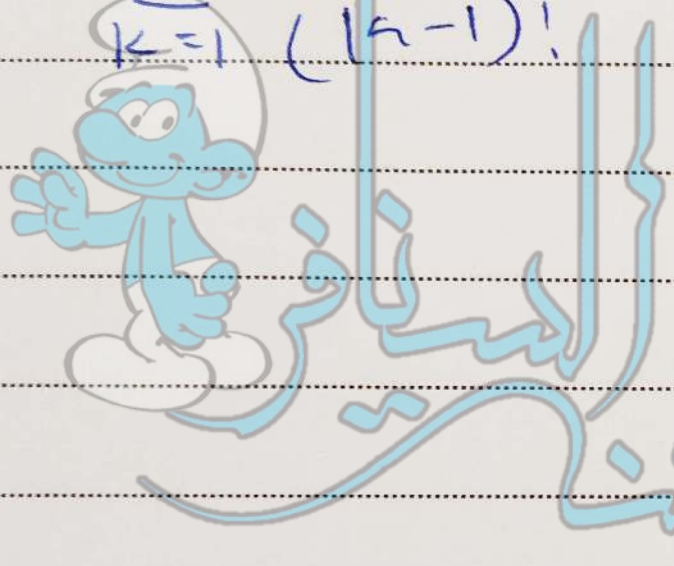
(A)

Subject / /

18) $\sum_{k=1}^{\infty} \left(\frac{aK+1}{2K+7} \right)^{K}$ Root test

$0 < \lim_{K \to \infty} \left(\frac{aK+1}{2K+1} \right)^{K/K} < 1$ عشان تكون conv.

$0 < \frac{a}{2} < 1 \longrightarrow 0 < a < 2$ (a)



14) $\sum_{k=1}^{\infty} \frac{2^{2k+1}}{(k-1)!}$ $\xrightarrow{k-1=n}$ $\sum_{n=0}^{\infty} \frac{2^{2n+3}}{n!}$

$2(k-1) = 2n$

$2k-2 = 2n$

$2k = 2n+2$

$2k+1 = 2n+3$

$2^3 \sum_{n=0}^{\infty} \frac{2^{2n}}{n!}$

تذكر: $e^x = \sum \frac{(x)^n}{n!}$

$8 \sum_{n=0}^{\infty} \frac{4^n}{n!}$ $\xleftarrow{\text{وازن}}$ $8e^4$

[C]

20) $\int \frac{1}{1+x^2} = \int \sum x^{2k} (-1)^k$

$$\tan^{-1} x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{2k+1}$$

$$\tan^{-1}(1) = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \sum_{k=0}^{\infty} \frac{(-1)^{1k}}{2k+1} + 1$$

بس هو مبلش على
$k=1$
و عازم على $k=0$

(B)

$$\frac{\pi}{4} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} + 1$$

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4} - 1$$

(B)